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BASECAMP · FUNDAMENTALS

Supervised vs unsupervised vs reinforcement learning

Supervised vs unsupervised vs reinforcement learning

There are three main ways to teach a machine, and they map surprisingly well onto how humans learn too.

- ◆ Supervised learning: learning from labeled examples with a known right answer. Like flashcards — "here's the input, here's the correct output, learn the pattern." Used for spam detection, price prediction, medical image diagnosis.
- ◆ Unsupervised learning: finding structure in data with no right answers given. Like being handed a messy pile of photos and asked to sort them into groups without being told what the groups are. Used for customer segmentation, anomaly detection.
- ◆ Reinforcement learning: learning through trial, error, and reward — no labeled data at all, just a signal for "good" or "bad" outcomes. Like training a dog with treats. Used for game-playing AI, robotics, and — this part surprises people — a key step in training chatbots to be helpful (RLHF: [Reinforcement Learning from Human Feedback](#)).

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“Which of these three feels most intuitive to you, and which still feels like a black box?”

Follow along – Day 5 drops tomorrow.

#MachineLearning

#ReinforcementLearning

#AI

#DataScience